FROM MASSES TO AGES

Kaelan Geng

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Supervisor of Dissertation

Christian Ginski, Full Faculty Title and Affiliation

Graduate Group Chairperson

Graduate Group Chairperson's Typed Name, Full Faculty Title and Affiliation

Dissertation Committee

Committee Member's Typed Name, Full Faculty Title and Affiliation
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ABSTRACT

FROM MASSES TO AGES

Kaelan Geng

Christian Ginski

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CHAPTER 1

Introduction

1.1. Motivation

Determining the age of a planet helps us ascertain the formation delay of the planet, by comparing it to the age of the parent star. We can also test our current formation models, and compare it to the expected disk lifetime (a few Myr, Strom et al., 1993, Manara et al., 2023): if the ages of planets are often younger than the disk lifetimes, then it suggests that disks are being replenished, perhaps through late infall (Kuffmeier et al., 2026). In addition, it helps constrain evolutionary models, assuming we have the planet's mass and luminosity. Normally, we infer a mass estimate from the luminosity and age, while assuming coevality; if we have all 3, we could constrain our evolutionary models accurately enough to determine if the cooling tracks we have are correct.

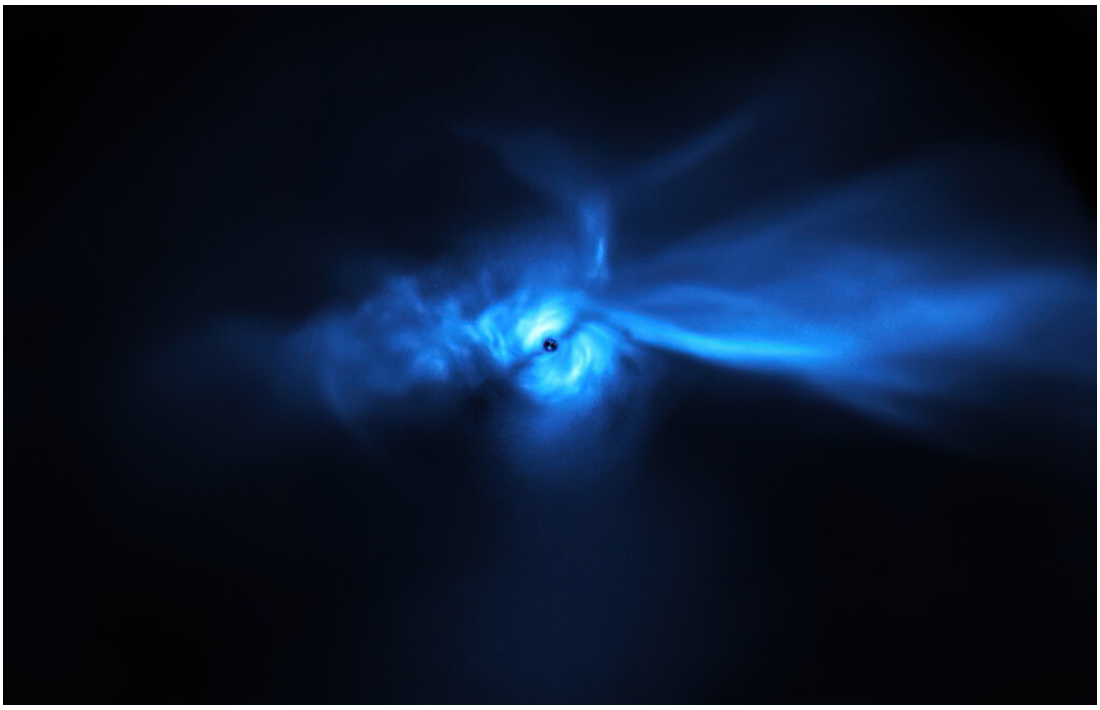


Figure 1.1: The SU Aurigae system. The majority of this photo comprises of wispy arms, which Ginski et al., 2021 theorise is material from the surrounding medium replenishing the material in the disk. Credit: Ginski et al., 2021

One of the major caveats to our current models are that planets and their parent stars are not exactly coeval (Bowler, 2016). Planets can span anywhere from the full stellar age, to 10 Myr younger, depending on the formation mechanism; top-down mechanisms would be close to coeval, while bottom-up mechanisms would leave a discrepancy of a few Myr. Furthermore, depending on the formation mechanism, it could lead to different inferred masses: take for example the young, planetary-mass companion 2M1207-3932 b of the TW Hydrae Association. Assuming coevality, its hot-start mass is $\sim 5M_{Jup}$. But, if the formation is delayed by 8 Myr, then the mass is only $\sim 2.5M_{Jup}$ (Bowler, 2016). Thus,

1.2. Formation Mechanisms

1.2.1. Core Accretion

The standard model for gas-giant formation by core accretion comes from Pollack et al. (1996), who built on older accretion models (Safronov, 1969; Mizuno et al., 1978; Mizuno, 1980; Bodenheimer and Pollack, 1986). A general overview of the process is as follows: a protoplanetary disk consists of dust and gas. This dust undergoes random motion, and eventually dust particles may collide and clump together. The repetition of this process eventually leads to pebbles (dust coagulation, Drążkowska et al., 2023), then boulders, plentesimals, then finally, through planetesimal accretion (Safronov, 1969), a planetary core massive enough to start accreting gas from the disk, which eventually leads to runaway gas accretion (Mizuno, 1980, Bodenheimer and Pollack, 1986, Pollack et al., 1996).

All in all, this process' timescale is on the scale of Myr (Pollack et al., 1996), which is significant because a planet formed in this way would be sizeably younger than its host star; planets formed through CA are also what this project aims to find.

This process occurs closer to the centre of the disk than gravitational instability, the other widely-accepted formation pathway, detailed in the next section. Note that this process is not necessarily linear; most pebbles actually don't evolve to the metre-scale. Among the potential barriers include the fragmentation-barrier, drift-barrier and others that are beyond the scope of this project but are

detailed in Drazkowska et al. 2023.

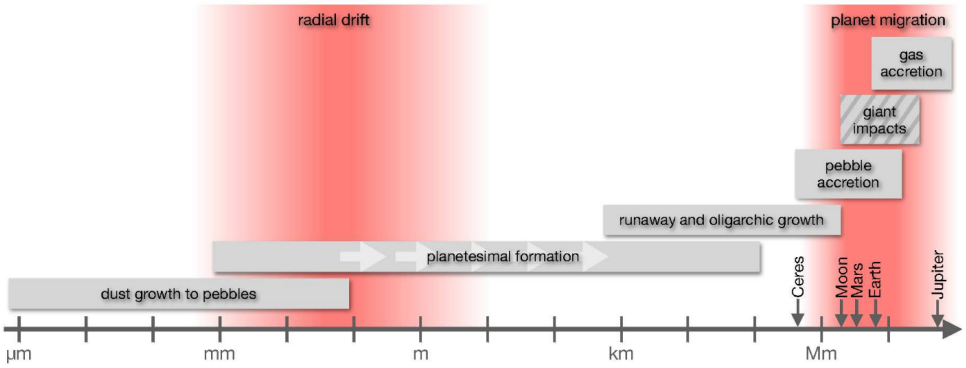


Figure 1.2: Fig. 3 from Drazkowska2023. This figure provides a general scale for the stages of CA.

Previous models that Pollack et al., 1996 model built upon include Mizuno et al., 1978 and Mizuno, 1980 of core instability, Bodenheimer and Pollack, 1986's investigations into Mizuno's models, Pollack et al., 1986 and Podolak et al., 1988 examinations into planetesimal accretion, among others.

In short, a critical mass before runaway gas accretion occurs is found to be $\sim 10M_{\oplus}$ (Mizuno, 1980), later $\sim 10M_{\oplus} - 30M_{\oplus}$ (BP86), and also when the heavy-element mass and the gas mass of the planet would equal; after this critical mass is reached, runaway gas accretion occurs. However, observations of the giant planets also revealed that Neptune and Uranus had compositions such that, in these models, their mass fractions were comprised of too little gas mass, so they'd have never reached the equality required to reach the critical mass. Pollack et al., 1996's remedy to this issue will be found in the outline below. Note also, that this model occurs after dust and pebble accretion and

Arranged neatly, the model of Pollack et al., 1996 is split into three phases: during phase 1, planetesimal accretion exceeds gas accretion rate; they're initially many orders of magnitude apart, with planetesimal accretion rate being the greater value. Overtime, the two rates become comparable: after $\sim 10^5$ years, they are comparable and planetesimal accretion rate begins declining sharply. Meanwhile, gas accretion rate keeps growing. This phase denotes a runaway planetesimal accre-

tion, separate from the runaway gas accretion, and it finishes when a protoplanet has emptied its feeding zone of planetesimals. However, the feeding zone grows as the planet's total mass increases, introducing new planetesimals (Pollack et al., 1996)

Phase 2 is a novel phase introduced in this model, that denotes a long, liminal phase where both accretion rates are relatively time-invariant, now with gas accretion rate exceeding planetesimal accretion rate. This phase spans the bulk of the aforementioned Myr timescale, with Pollack's simulations of Jupiter lasting $\sim 7Myr$. Finally, phase 3 denotes the runaway gas accretion, where both accretion rates (with gas still far exceeding planetesimal) start rapidly increasing again.

Fig 1c of Pollack et al., 1996 is particluarly relevant to this project: during phase 3, the luminosity grows rapidly to a stage that we can image directly. Once it reaches a maximum luminosity, it will cool down for the rest of its lifetime (since planets have no mechanisms that illuminate them); this cooling is what this project will utilise to infer the age from the luminosity and dynamical masses.

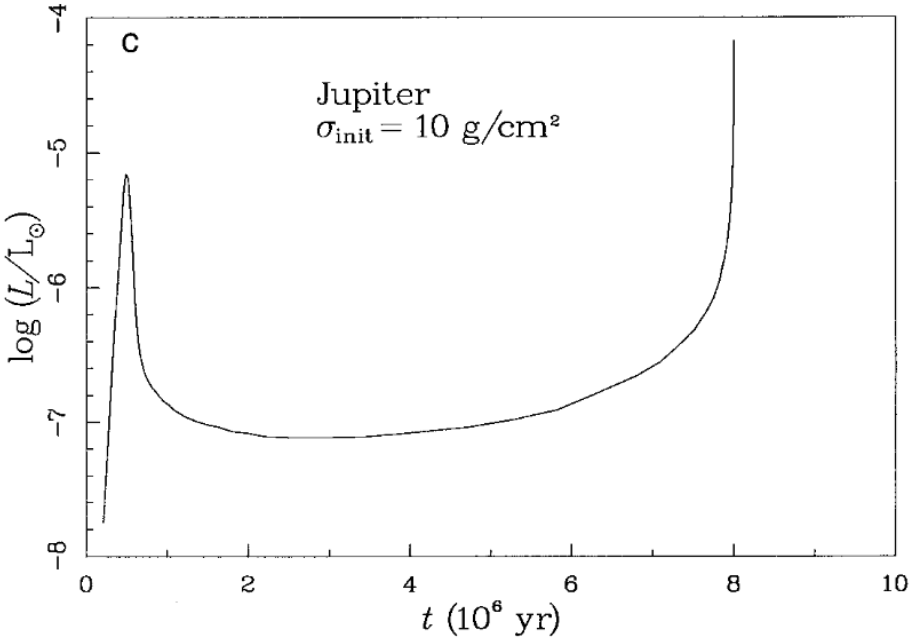


Figure 1.3: Fig 1c of Pollack et al., 1996. The initial rise corresponds to the runaway planetesimal accretion of phase 1; the long plateau corresponds to the transitionary phase 2; the sharp increasing corresponds to both accretion rates rapidly increasing (and the runaway gas accretion) in phase 3

1.2.2. Gravitational Instability

1.2.3. A Note on Cloud Fragmentation and Brown Dwarfs

Cloud fragmentation is traditionally a mechanism that forms stars. A general overview (see Whitworth et al., 2006 for the full process) is that given a molecular cloud is massive enough, it contracts and fragments into subclouds. This process repeats for the subclouds, until eventually the subclouds are so optically thick that they can't contract anymore since they can't radiate away the contraction heat (Whitworth et al., 2006).

The floor of the mass of objects formed seems to reach well into the planetary-mass range ($\sim$ 1 to 4 M_{Jup}, Whitworth and Stamatellos, 2006). Also traditionally, the criterion for brown dwarfs is that it exceeds the deterium burning limit of $\simeq 13 M_{Jup}$ (Chabrier and Baraffe, 2000), however even this remains challenged; Chabrier et al., 2014 argues plantary and brown dwarf can overlap and that there is no clear cut-off mass for one body to be a brown dwarf and another a planet.

Actually, the criterion for what a planet actually *is is even under debate:*

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Furthermore, the formation tracks for brown dwarfs are also contested: many studies theorise that brown dwarfs can form from the low-end masses of cloud fragmentation (Balmer et al., 2024, Chabrier, 2003). Some others suggest they could also be formed through disk instability (Boss, 1997, Stamatellos et al., 2007, Stamatellos and Whitworth, 2009) or even core accretion

along with the associated runaway gas accretion (Balmer et al., 2024, D'Angelo et al., 2010, Mollière and Mordasini, 2012).

Thus, this project will also make use of lower mass brown dwarfs under the assumption that they can use the same formation tracks as planets.

1.3. Imaging

1.4. Evolutionary Models

Evolutionary models are split into three categories: hot-start, warm-start and cold-start models. Hot-start model planets start with high entropy retain high entropy after formation, resulting in a large initial radius and luminosity (Bowler, 2016, Bussmann and Helled, 2026). Cold-start models were made prominent by direct imaging to emulate more realistic scenarios for giant planets formed through CA (Bowler, 2016).

Traditionally, cold-start models are associated with core accretion and hot-start models with disk instability. However, there is a bit more nuance: core accretion can also produce warm or hot initial conditions

CHAPTER 2

Methodology

This project makes use of the species Python package (Stolker et al., 2020). The cooling-curve inversion uses a brute force technique, by using Monte Carlo to sample within the uncertainties of the parameters that are passed in.

2.1. Figures

The captions of figures inside the figure environment (see Figure **??**) will display in the List of Illustrations. Use the `\label` to label figures.[1]

[1]This also includes photographs, maps, schemes, diagrams, etc.

APPENDIX A

TITLE OF APPENDIX A

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A.1. Section Name of Appendix A

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APPENDIX B

TITLE OF APPENDIX B

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